The Short, the Long, and the Consequences of VAR Analysis for Structural DSGE Estimation *

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Abstract

De Graeve and Westermark (2025) advocate the use of long lags VARs for structural analysis, claiming that the truncation bias is a severe problem for the IRFs estimation. VARs can be used to create additional moments to estimate DSGE models minimizing the distance between their IRFs and therfore estimating the underlying structural parameters. In this study we ask what are the consequencies of enriching the lag structure of a VAR used for DSGE estimation. I replicate the results of Altig et al. (2011) and I find that the both IRFs and Structural parameters are sensible to the chosen lag lenght. Matching the DSGE with a VAR(12) instead of the baseline VAR(4), I find that the Calvo Price parameters that differencieate the observationally equivalent Altig et al. (2011) model between firm specific capital vs homogeneous capital are no longer statistically different from each other. The implied duration of the Calvo Price parameter is 2.5 times higher than the baseline, re opening the micro-macro discrepancy in price stickiness.

Keywords: Truncation Bias, Long Lag VARs, DSGE Estimation, Minimum Distance, IRFs
JEL Codes: C32, C52, E32

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1 Introduction

1.1 Motivation

Dynamic Stochastic General Equilibrium (DSGE) models can be mapped in the form of State Space representation, which then can be retrived as Vector AutoRegressions (VAR) of infinite order. Impulse Response Function Matching Estimation (IRFME) is a limited informaiton approach that minimises the distance between VARs' IRFs and the one implied by a DSGE model. This technique requires that sample and theoretical IRFs are identified by restriction implied by the DSGE model, which requires a SVAR representation, so that the IRFs are informative fro the DSGE parameters Rossi et al. (2012). If the reducced form VAR (RF-VAR) adequately captures the Data Generating Process (DGP), IRFs are optimal at each horizon. But if the RF-VAR is misspecified, the specification error naturally spill to strucutral statistics and will be compounded at each horizon, Ramey (2016); Canova and Fosso (2025).
Since is not feasible to estimate a VAR(∞) as samples are finite, researchers truncate the VAR which becomes misspecified, and more mispecified the smaller the chosen lag lenght De Graeve and Westermark (2025); Barnichon et al. (2026) "all VAR researchers should be concerned about lag truncation, regardless of the horizon of interest dictated by the research question. Truncation affects inference at all horizons, including the very short ones ($h \leq p$)" De Graeve and Westermark (2025) "The overarching implication of our investigation is that, contrary to conventional wisdom, long-lag VARs are feasible in practice. Applied research should use them broadly because they safeguard against the significant problem that lag truncation implies for structural economic conclusions." De Graeve and Westermark (2025). "it is up to the researcher to show that her structural results are robust to significant increases in the lag length, far beyond the lag lengths typically considered in current practice." De Graeve and Westermark (2025)

1.2 Contribution and Results

I analyse the consequences of enriching the lag structure of a VAR used to estiamte a DSGE with IRFs minimum distance estimation. I replciate Altig et al. (2011) and find that the both IRFs and Structural parameters are sensible to the chosen lag lenght.

The Calvo Price parameters that differenciate the observationally equivalent Altig et al. (2011) model between firm specific capital vs homogeneous capital are no longer statistically different from each other. This go against their main conclusion that the firm specific capital model is the one that better fit the data.

(Maybe the Capital Adjustment Cost variable that since Altig et al. (2011) researchers have started to use, is not fundamental to explain inflation inertia.)

1.3 Related Literature

I contribute in XXX breanches of research. The first is the one that study ... Long Lag VARs De Graeve and Westermark (2025); Baumeister (2025); Antolín-Díaz and Surico (2025); Antolín-Díaz et al. (2026)
The second branch lies between the fast growing debate between Local Projections and VARs IRFs (VARs and LPs) Ludwig (2026); Li et al. (2024); Òscar Jordà (2005)
State Space rappresentation of DSGE Ravenna (2007); Fernández-Villaverde et al. (2005); Forni et al. (2016); Theodoridis (2011); Giacomini (2013)

1.4 Outline

The rest of the work is organised as follows. In Section 2 I present the Truncation Bias problem, the theory that maps DSGE into VAR(inf) that suggests to increase the lag length. In section 3 I summaryse briefly the Altig et al. (2011) DSGE. In section 4 I present the empirical results and in section 5 I conclude.

2 Truncation Bias: Illustration of the Problem

VARs has at least three source of bias: Small Sample Bias (cit uno degli anni 50, l'ho visto penso su Chari et al 2008, anche loro li posso citare qua), Identification Bias Forni et al. (2016), and Truncation Bias De Graeve and Westermark (2025); Baumeister (2025); Barnichon et al. (2026); Chari et al. (2008); Ravenna (2007), which is the principal focus of this work.

To explain it, I follow the single equation and then generalization to a VAR example used by De Graeve and Westermark (2025).
Let's consider a DGP

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t \tag{1}$$

where the variable y_t is determined by two exogenous variables, x_{1t} and x_{2t} and a shock ϵ_t. If y_t is estimated with a regression omitting one of the two exogenous variables, say x_{2t},

$$y_t = x_{1t}b_1 + e_t \tag{2}$$

the omission leads to biaesd point estimate (unless $x_{1t} \perp x_{2t}$).

$$\mathbb{E}(b_1) \neq \beta_1 \tag{3}$$

with an always upwardly biased variance estimate.

$$s^2 > \sigma^2 \tag{4}$$

Translating this to VARs

$$Y_t = B_1 Y_{t-1} + ... + B_p Y_{t-p} + U_t, \mathbb{E}[U_t U_t'] = \Sigma_U \tag{5}$$

$$B(L) = B_1 L + ... + B_p L^p \tag{6}$$

which has $p << p^*$, where p^* denotes the true unknown lag length, the VAR will suffer from truncation bias. Lag truncation cause bias in reduced form coefficients $B(L)$ and in the reduced form variance covariance matrix Σ_U. Which are both ingredients of IRFs

$$IRF = f(B(L); \Sigma_U) \tag{7}$$

If the components are biased by lag truncation, so will the IRFs. De Graeve and Westermark (2025) explores the trivial solution to truncation bias, that is the dramatic increase of lag length. They find evidence against the conventional wisdom that uncertainty would become pervasive as the lag choice would increase the number of parameters rapidly, hence reducing the degrees of freedom and making confidence interval width prohibitively large. But the standard intuition is only part of the story, because increasing the lag choice not only reduces the misspecification in the well known bias reduction, but also it

reduces the variance. This reduction in variance works against the imprecision resulting from increased parametrization De Graeve and Westermark (2025).

3 DSGE State Space Representation and VARMA

Simple theoretical derivation describing that even if the Altig et al. (2011) is stochastic singular (and defend the fact that most of the SVAR analysis is done with stochastic singular models (more observables than shocks), Forni et al. (2016)) and explain that the solution of the DSGE in State Space can be represented as a VARMA Ravenna (2007), hence the use of VAR with long lags is justified even if the ABC and D Fernández-Villaverde et al. (2005) representation of the VAR(∞) does not exist.

4 Theoretical Model

Then a breif description of the Altig et al. (2011) DSGE using the replication package from (ALEL NBER 2005 WORKING PAPER)

5 Empirical Results

5.1 Implied Duration from Calvo Price Table: lags 4 8 12

Table 1: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR ($p = 4$)**: Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

6 Conclusion

References

Altig, D., Christiano, L. J., Eichenbaum, M., and Lindé, J. (2011). Firm-specific capital, nominal rigidities and the business cycle. *Review of Economic Dynamics*, 14(2):225–247.

Antolín-Díaz, J., Fernández-Villaverde, J., and Guerrón-Quintana, P. (2026). Double descent in time series. Working paper, MIT, University of Pennsylvania, and Boston College.

Antolín-Díaz, J. and Surico, P. (2025). The long-run effects of government spending. *American Economic Review*, 115(7):2376–2413.

Barnichon, R., Mesters, G., Nakamura, E., and Steinsson, J. (2026). Humble vars. Working paper, University of California, Berkeley.

Baumeister, C. (2025). Discussion of "local projections or vars? a primer for macroeconomists". *NBER Working Paper Series*.

Canova, F. and Fosso, L. (2025). Low frequency movements and svar analyses. CAMP Working Paper 10/2025, Centre for Applied Macroeconomics and Commodity Prices (CAMP).

Chari, V. V., Kehoe, P. J., and McGrattan, E. R. (2008). Are structural vars with long-run restrictions useful in developing business cycle theory? *Journal of Monetary Economics*, 55(8):1337–1352.

De Graeve, F. and Westermark, A. (2025). Long-lag vars. *Journal of Monetary Economics*, 156.

Fernández-Villaverde, J., Rubio-Ramírez, J., and Sargent, T. J. (2005). A, b, c's (and d)'s for understanding vars. NBER Technical Working Paper 308, National Bureau of Economic Research.

Forni, M., Gambetti, L., and Sala, L. (2016). Var information and the empirical validation of dsge models. RECent Working Paper 119, RECent - Center for Economic Research, University of Modena and Reggio Emilia.

Giacomini, R. (2013). The relationship between dsge and var models. cemmap Working Paper CWP21/13, Centre for Microdata Methods and Practice (cemmap).

Li, D., Plagborg-Møller, M., and Wolf, C. K. (2024). Local projections vs. vars: Lessons from thousands of dgps. Working paper, Two Sigma Investments, Princeton University, and MIT.

Ludwig, J. F. (2026). Local projections are var predictions of increasing order. *Working Paper, R&R Econometrica 2nd Round*.

Ramey, V. A. (2016). Macroeconomic shocks and their propagation. In *Handbook of Macroeconomics*, volume 2, pages 71–162. Elsevier.

Ravenna, F. (2007). Vector autoregressions and reduced form representations of dsge models. *Journal of Monetary Economics*, 54(7):2048–2064.

Rossi, B., Hall, A. R., Inoue, A., and Nason, J. M. (2012). Information criteria for impulse response function matching estimation of dsge models. *Journal of Econometrics*, 170(2):499–518.

Theodoridis, K. (2011). An efficient minimum distance estimator for dsge models. Bank of England Working Paper 439, Bank of England.

Òscar Jordà (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

A Replication Files and Code

Detail where the reader can find your code (e.g., a GitHub link).

B Additional Tables

C Additional Figures

C.1 ACEL 2005 VAR Variables

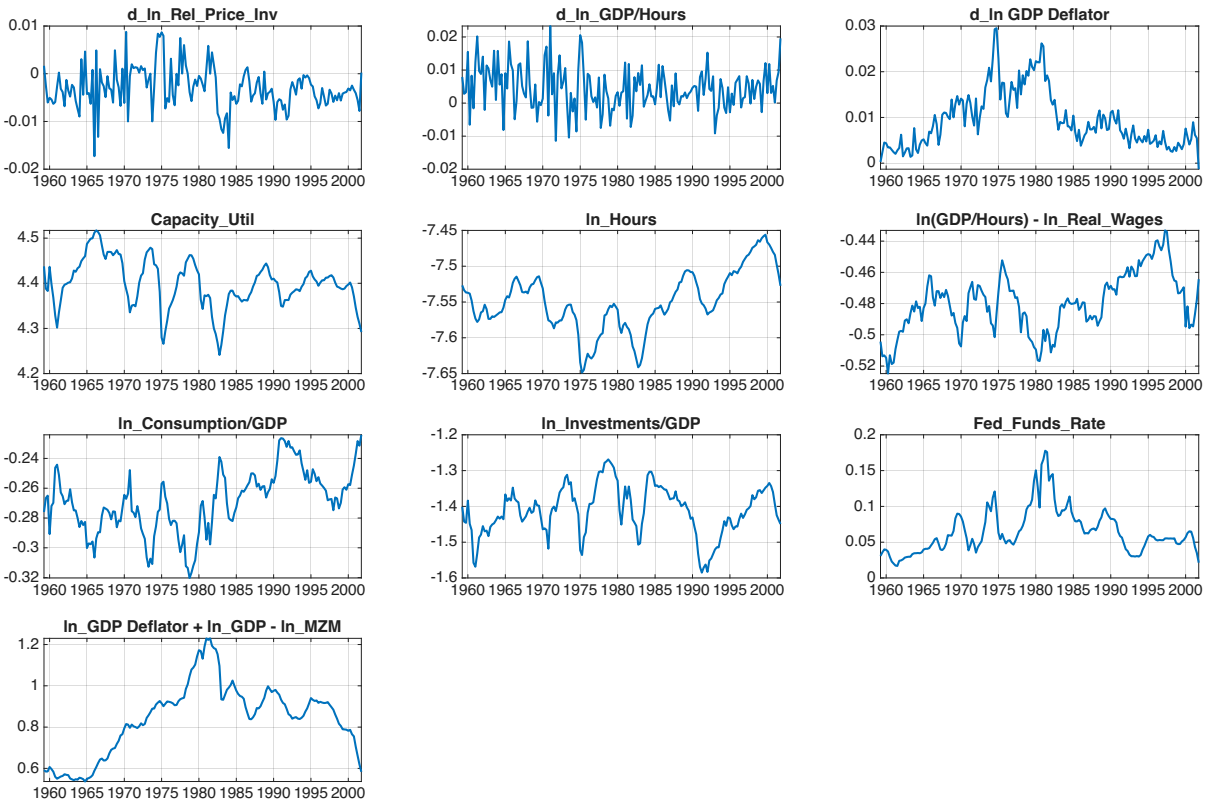


Figure 1: Raw Historical Series